The pxCORE package

Core aggregation and key management for P3CTeX

Scvria
P3CTeX / UOC PEC Repository

2026/02/26 – Version 0.2

Abstract

pxCORE is the central aggregation package in the P3CTeX stack. It owns the pxCORE key family, which controls activation of standalone libraries (pxPRP, pxUML, pxSRC, ...) and offers a minimal LaTeX2e configuration API.

Contents

1 Loading the package **1**

2 Readable tool selection **1**

3 Key family pxCORE **2**

4 User command reference **2**
4.1 \pxCOREsetup 2
4.2 \pxCOREIfDebugTF 3

5 Minimal example **3**

1 Loading the package

In standalone documents, load pxCORE as a normal package:

```
\usepackage[<options>]{pxCORE}
```

In P3CTeX-based workflows, pxCORE is normally loaded by the document class (P3CTeX.cls) and configured via class options.

2 Readable tool selection

```
\usepackage[code,images,tables,diagrams,calculations,requirements]{pxCORE}
```

The keywords `code`, `images`, `tables` and `diagrams` select the existing LST, SRC, TAB and CMP tools. `diagrams-config` forwards diagram options, just like `CMPconfig`. `calculations` (or `CALC`) loads `pxCALC`; `requirements` (or `REQ`) loads `pxREQ`. The diagram entry point `pxDIAG` delegates to `pxCMP`. All selection keys accept `=false`; a bare keyword means true. See the student-tools guide for examples and customisation.

3 Key family pxCORE

The `pxCORE` key family controls which libraries are activated and stores a few global flags:

`debug` (bool) enables internal debugging. When set, `\pxCOREIfDebugTF` will take its "true" branch.

`print-doc` (bool) reserved flag for future "print documentation" workflows. Currently stored but not interpreted.

`PRP` (bool) toggles loading of the `pxPRP` property-map library.

`UML` (bool) toggles loading of the `pxUML` UML diagram library.

`UMLconfig` (token list) raw option string passed to `pxUML`. When non-empty, `pxCORE` executes

```
\RequirePackage[<UMLconfig>]{pxUML}
```

instead of a bare `\RequirePackage{pxUML}`.

`SRC` (bool) toggles loading of the `pxSRC` included-sources library.

Keys are typically set via package options:

```
\usepackage[PRP,UML,SRC]{pxCORE}
```

or adjusted later with `\pxCOREsetup` (see below).

4 User command reference

4.1 \pxCOREsetup

Synopsis

```
\pxCOREsetup{<key>=<value>,...}
```

Updates the pxCORE key family after package loading. This is useful to toggle flags such as debug programmatically or to adjust options in local scopes.

Example:

```
\pxCOREsetup{debug=true}
...
\pxCOREsetup{debug=false}
```

4.2 \pxCOREIfDebugTF

Synopsis

```
\pxCOREIfDebugTF{<true-code>}{<false-code>}
```

Branch on the debug flag in the pxCORE key family. Both branches are arbitrary LaTeX code.

Example:

```
\pxCOREIfDebugTF
  {\typeout{[pxCORE] Debug mode enabled}}
  {}
```

5 Minimal example

```
\usepackage[PRP,UML,SRC]{pxCORE}
\begin{document}
\pxCOREsetup{debug=true}
\pxCOREIfDebugTF{[DEBUG-ON]}{[DEBUG-OFF]}
\end{document}
```